NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.4c Thunderstorms: AI forecasts driving WRF

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 14:38 — 2026-11-13 09:00

Stage conclusion

Project report

Thunderstorms: AI forecasts driving WRF (K)

Goal

Using the same WRF setup (4 km single domain, no cumulus parameterization), drive initial and boundary conditions separately with Pangu forecast fields and GFS forecast fields, initialized 24/48/72 h ahead for thunderstorm events that have already been simulated and have observations; compare storm location and timing, radar objects and station gusts to judge whether Pangu-driven runs are no worse than GFS-driven runs. Serves as the accuracy yardstick for the L nodes.

Current status

10-24 check: the R-group comparison (option ① of A173, approved by the PI on 10-21) has been fully run and finalized; no jobs are running. - **Conclusion v2**: , six-reviewer review 6/6 valid, finalized under reading A. - For the single Akron event and this configuration, Pangu 13-level forcing did not reach "no worse than GFS forcing with the same number of levels" at any of the three lead times 24/48/72 h (3/3). - The number of levels (including the Q2 re-derivation) was judged only at 24 h, and only at the margin: the position exceeded the 50 km tolerance by 2.8 km. At 48 and 72 h, timing cannot be judged and the factors cannot be separated. - Do not write "AI is worse than GFS." - **Stop category**: remains blocked (awaiting the PI). Remaining options for the PI to decide: ② extend to ≥3 events, ③ find an AI model with full-level output, ④ invest no further (see). - Decision record: .

Decisions made

- 2026-10-21/22 time-step unification: because R48 crashed at dt20, 48 h switched entirely to dt15 (G48/P48 rerun); because R72 crashed at dt15, 72 h switched entirely to dt12 (G72/P72 rerun); original time steps are kept as supplementary columns, and inconsistent judgments are recorded as "time-step sensitive · not judged"; if R72 crashes again at dt12, 72 h is "not separable" → last two sections
- 2026-10-21 the PI approved option ① of A173 (R-group 13-level GFS comparison) →; experiment card (criteria locked 10-21 09:45)
- 2026-10-17 shelved (did not reach "worth continuing"): 0/3 no worse than at 24/48/72 h, both groups' 24 h positions >150 km; do not write "AI forcing is worse than GFS" (model and number of levels not decoupled, R24 not run); restart conditions in decision →
- 2026-10-15 definition addendum (design gap, criteria unchanged): minimum area of ≥40 dBZ objects formally ≥64 km^2 (none/≥16 km^2 reported in parallel; if the three tiers disagree, that lead time is marked sensitive and not counted toward "worth continuing"); the timing-metric search window start moved earlier to 00Z (position window still 03–09Z); if an object already exists in the first 00Z frame, the timing metric is recorded as left-censored and not judged →; v2 revised per review : votes count only when the two tiers (16/64) agree, and neither positive nor negative votes count when they disagree; the threshold applies to all object judgments, and "within R km" uses the nearest grid point; left-censored cases substitute the interval into the original formula, and if not judgeable no positive vote is counted (ruling)
- 2026-10-14 design v2: GFS group changed to full levels (Deng 2025 abstract: GFS cut to 13 levels is the worst); 13-level GFS downgraded to optional R24 →
- 2026-10-14 experiment design v1: Pangu main experiment uses GFS analysis initial conditions (same initial conditions as the GFS group); ERA5-initialized runs only as optional E24; both groups use the same 13 levels / 6 h / same land surface; 6 h anchor pair and gate added →
- 2026-10-14 opened; first event Akron 2020-06-09, Wister 2018-08-10 only as a triggering comparison; WRF on CRC (no SU charged) or a reduced domain →

Tried and results

Date	What was done	One-line result
2026-10-22	48 h formal pairing dt15 (WRF line 1611124) + supplementary G48/P48 dt20; lead judged with judge_r48.py	(a) R48 vs G48: position 79.8 ≤ 379.6 ✓, timing R left-censored [190,∞) vs 230 → not judgeable → not judged; (b) P48 vs R48: position 206.9 > 129.8 → not no-worse (formal/supplementary and both tiers agree) → 48 h not separable
2026-10-22	R24 integration (CRC 1602758, dt20) + three tables (1607937/1608940, G24 regression 1608439 identical cell by cell); lead judged per card ⑤	(a) R24 vs G24: position 236.2 > 233.4 (exceeds by 2.8 km) *X*, timing 0 ≤ 110 ✓ → not no-worse (both tiers agree); (b) P24 vs R24: 64 tier position 307.9 > 286.2 and timing 170 > 60 *X*, 16 tier timing ≥160 > 90 *X* → not no-worse; R24 maximum 10 m wind within 100 km 20.5 m/s (0.55 of the observed gust 37.04)
2026-10-21	wx generated 13-level GFS forcing fields (CRC 1601835)	30 files, SHA manifest …, lead fully checked at 14:27, read-back PASS
2026-10-17	WRF line delivered the 72 h pair (formal G72/P72 dt15, CRC 1571882; dt20 supplementary 1560832) and filled in 150 km existence; lead counted votes	72 h, all four judgments agree: P72 position 307.7/299.7 km vs G72 104 km, at the 64 tier G has an object and P does not → not no-worse; all three lead times 24/48/72 counted as "clearly worse" (48 h on timing only); per the conclusion rules, proposed "shelved (AI forcing worse than GFS, single-event qualitative)"; sent to 6 reviewers, status unchanged before the ruling
2026-10-15	WRF line delivered G48/P48 formal v2 tables (job 1552228)	≥64 km²: G48 first appearance 00:30Z (−160 min), 06:10Z centroid error 316.8 km, object 80 km², maximum wind 17.1; P48 no first appearance within 50 km in the window (right-censored), error 273.3 km, object 287 km², maximum wind 17.0; ≥16 tier G48 −140 min / 275.2 km, P48 no first appearance / 174.7 km. Lead checked the CSV. Requested an added column for 150 km existence (each pair); judgment awaits the 72 h pair (about 10-18)
2026-10-15	WRF line accepted the pairing arrangement	Formal 72 h pair = G72 dt15 vs P72 dt15 (1547482 running); G72 dt15 new root on CRC, to pass the WRF line review before submission, expected to finish the evening of 10-17; G72 dt20 (1544570) as supplementary; 48 h pair delivered the evening of 10-15

Date	What was done	One-line result
2026-10-15	WRF line delivered G24/P24 formal v2 tables (job 1548771)	≥64 km²: first appearance observed 03:10Z, G24 04:00Z (+50 min), P24 00:20Z (−170 min); 06:10Z centroid error G24 183.4 / P24 307.9 km; maximum 10 m wind 22.2 / 21.5 m/s. ≥16 tier: P24 left-censored (first 00Z frame), another 32 km² object selected, position 233.1 km. Lead checked the CSV. Judgment awaits all three being aligned; for the 72 h pair, P72 switched to dt=15 s because of CFL, and G72 has been asked to rerun with the same dt
2026-10-15	WRF line delivered G06/P06 formal v2 tables (job 1545333)	≥64 km²: first appearance observed 03:10Z, G06 04:40Z (+90 min), P06 00:50Z (−140 min, within spin-up, anchor only); 06:10Z centroid error G06 53.1 / P06 66.6 km; object area observed 1380, G06 928, P06 1055 km²; maximum 10 m wind within 100 km 24.8 / 26.7 m/s vs observed gust 37.0. G06 gate passed under v2 (a ≥64 km² object within 150 km). ≥16 tier: P06 selected another 48 km² object. N0Q cross-check: positions overlap, area magnitudes differ; used only as a check with stated limitations. Lead accepted; definitions unchanged
2026-10-15	WRF line adopted definition v2	New script version (old version kept) judges hit50/150 km/gate after area filtering; G06 gate rechecked at ≥64 km²; formal tables first for G06/P06, paths reported for each remaining pair, longest group finishes about the morning of 10-16; fallback follow-up check at 10-16 16:41 still scheduled
2026-10-15	WRF line adopted two definition changes	Formal tables: 3 area tiers × 2 window starts; G06/P06 first, the other 6 integrations finish at the latest about the morning of 10-16; the observed ≥40 dBZ object at 06:10 is only one 2 km grid point, suspected error in the three-radar composite; being compared against IEM N0Q (if confirmed, under the formal definition the observations may have no object, which must be flagged as is for the lead to judge)

Date	What was done	One-line result
2026-10-15	ideas generated forcing fields v2.2 (CRC job 1544281, CO-, 0 SU)	All 66 file SHAs passed (lead rechecked); only Q2 changed, everything else byte-identical to v2.1; within 300 km of KAKO, new minus old −0.37 to −0.91 g/kg, exactly the opposite sign of the WRF line report, i.e., equal to the reference algorithm
2026-10-14	WRF line G06/P06 preprocessing trial (jobs 1542899/1542900)	metgrid passed (G 32 levels / P 14 levels); sea-surface SNOW=0, SMOIS=1, SST=TSK confirmed; real.exe for both groups produced NaN in vertical interpolation in the boundary zone at 06-09 12Z (WRF line judged it a problem in their pipeline); Q2 sensitivity: v2.1 too moist by +0.38 to +0.91 g/kg, maximum absolute difference 1.6 to 9.3 g/kg, all above 1 g/kg → changed to v2.2
2026-10-14	ideas generated 8 sets of forcing fields (v2.1, 0.25° GFS)	66 WPS files, 1.6 GB, read-back and SHA checks passed (lead spot-checked 4, OK); G 31 levels / P 13 levels, land surface byte-identical between the two groups; sea-surface soil and snow filled from the nearest grid cell, Pangu negative q clipped to 0 (both marked as assumptions)
2026-10-14	ideas forcing-field pre-check	2020 GFS 0.5° multi-lead-time forecasts no longer have a public archive (NCEI 404, only about 2 years kept; AWS keeps only 30 days); native 0.25° downloadable from NCAR GDEX d084001; Colab not opened, 0 CU
2026-10-14	Grok quick search for precedents	No precedent for AI-driven CPM of mid-latitude downbursts/thunderstorm winds; Deng 2025 abstract checked: 33-level GFS ≈ 13-level Pangu, GFS cut to 13 levels is the worst; Brewer 2025 abstract checked: GraphCast-driven WRF comparable to GFS/ECMWF-driven (downslope winds)

Parameter source table

Parameter	Value	Source or "assumption"	Places checked
WRF d01 settings	4 km, 748×600×80, dt 20 s, cu0, mp8, PBL2/sf2, lw4	Author namelist	WRF lead reply

Parameter	Value	Source or "assumption"	Places checked
Compute time	About 1 SU per model hour (not benchmarked)	WRF lead converted from exp-A	Awaiting short test
Projection	Lambert, 38N/100W, truelat 38/38	WRF line; author wrfinput metadata	—
grid_fdda	0	Author namelist.input line 113	—
GFS source for this experiment	Native 0.25°, NCAR GDEX d084001	ideas pre-check (README)	NCEI, AWS and d335000 have no 2020 0.5° forecasts
RH→q (G group), Pangu negative q clipping, nearest-cell fill of sea-surface soil/snow	See	**Assumption** (implementation)	Awaiting metgrid/real test
2 m specific humidity	v2.2: q from the lowest complete pressure level with GHT>SOILHGT (v2.1 ln p interpolation voided)	**Assumption** (design v2.2; based on the WRF line Q2 sensitivity, no literature basis)	Brewer repository: its GFS does not provide 2 m RH
Author forcing	GFS 0.5°, 00Z initialization, 3 h	WRF line	—
Source of observed objects	NEXRAD Level-II column-maximum reflectivity composite + KAKO METAR/special reports; gust anchored at 37.04 m/s	WRF line 10-15 (ASOS 1-minute data missing for the day, no MRMS in the NCEP archive for the day)	Criterion thresholds unchanged
Observed peak time	About 06:10 UTC (KAKO 02:10 EDT)	GZYPXN4S PDF pp.9–13 (cited via WRF line REPORT_FINAL)	To be checked by this node
GFS yardstick number of levels	Full levels (not cut to 13)	Deng 2025 AOSL 10.1016/j.aosl.2025.100589 abstract (original abstract read via OpenAlex)	Full text not read (ScienceDirect 403)
Pangu initial conditions	GFS analysis at the same time	Not stated in the Deng 2025 abstract; Grok claims its SI states GFS analysis is used; **not checked**	Awaiting ideas to check the original
Surface/soil fill, 2 m humidity, PSFC	Land surface from GFS f000 at initialization; 2 m humidity derived the same way for both groups; PSFC not provided	**Assumption** plus WRF lead opinion; Brewer 2025 repository rfovell/GraphCast-WRF (Apache-2.0, last push 2024-12-19) has a fill approach to be checked	Delegated
WRF version	4.0.3	WRF lead (author results are version sensitive)	—

Open items and next steps

- ideas delivers forcing fields → check field list and parity → formally notify of the path.
- Grok quick search has returned (see Tried and results). Deep Research step 2 **not done**: 30-day quota 30/30 used up; to be sent once quota recovers or the PI adds quota.
- WRF lead accepted v2: the conclusion must state that the difference in the main comparison includes both model and number of levels; with spare capacity, run R24 first, then E24; one metgrid/real trial each for G06 and P06.
- Forcing fields v2.2 delivered (10-15 05:00 --report to the parent node, forwarded to the WRF lead via the root lead).
- WRF line rechecked v2.2 at 10-15 04:58 and started runs: the boundary NaN in real was bypassed with all_ic_times=true (measured to have bit-for-bit no effect on boundary forcing); G06/P06 run first (about half a day), and the G06 gate is judged per section 5; after the gate, the six integrations of the 24/48/72 pairs run in parallel (the longest, 78 h group about 16–18 h wall clock, queueing extra). 3D fields kept only for 03–09Z, target <15 GB. Numerical tables stored in and returned to this node; this node judges per the frozen criteria and reports to the parent node. Fallback follow-up check at 10-16 16:41.
- Waiting on : design objections, delivery location, schedule.
- WRF line short test → 06 pair passes gate → 24/48/72 pairs → comparison table → conclusion (before giving to the PI

Boundaries

- Rules of the testing ground: no per-experiment cap on Colab (the parent project corrected this on 10-14 per the PI's words "use Colab a lot at this stage, no CU limit"; the original ≤20 CU is voided); if a single experiment is expected to exceed 100 CU, stop first and report to the parent node lead; always colab stop after use and do not let sessions idle; dispatch only one executor per round; supercomputer single job ≤50 SU (CRC charges no SU); WRF is submitted by the WRF line bot and requires a ComputeOps resource request, registration, and pre-check and submission as two separate commands; other lines' data are read-only.
- Storage and compute location (tightened by the PI on 10-14): the workstation does not run computations or store computational data; it holds only scripts, small results <200 MB per item, figures and reports; if >1 GB must be stored temporarily, first report to the parent node lead. Forcing fields and WRF output go only on Colab or the clusters (currently CRC); new large data go preferably to TACC [local path omitted] or [local path omitted] (CRC group space about 95%); CRC defaults to a single node. Every task brief must state where to compute and where results go.
- Hierarchy (set by the root lead on 10-14, 5-level cap): this node is a second-level node, **no further third-level sub-projects**; when subdivision is needed, use tasks within this node, or report to the parent node lead (thunderstorms and convective scale) to open a sibling second-level node.
- Only two hops remain downward: this node → bot → tool. Task briefs do not allow tool bots (gpu-gemini, reviewers, litwiki, etc.) to delegate further.
- The subtree list of is not grouped by parent node; the project tree in the interface is authoritative.

Literature basis

- Akron mechanism: card 2DPBW6VX PDF pp.6, 9–11; observations: Childs 2021 card GZYPXN4S Table 2, PDF p.16 (cited via the WRF lead, to be checked by this node)
- Xu 2024 ERL 10.1088/1748-9326/ad41f0; Liu 2024 JGR:MLC 10.1029/2024JH000207 (no cards built, not in the local library)
- Deng 2025 AOSL 10.1016/j.aosl.2025.100589 (abstract checked, no card built); Brewer 2025 WAF 10.1175/WAF-D-24-0097.1 (abstract checked, no card built); these 4 papers have been delegated to litwiki to build cards and check the originals (10-14 report: PDFs not retrieved, passed to Literature to retrieve full texts, cards to be built once retrieved); not cited in conclusions until cards are built.

PI notes

- 10-14: fundamentally different in scale from tropical cyclones (B3); listed separately.

10-23 update: 72 h judgment and merged draft for the three lead times

- 72 h (formal dt12, WRF readout 1617717 pass_all, R72 dt12 did not crash): number-of-levels timing not judgeable (no first appearance within the window for all three groups) → not judged; model not no-worse (P72 290.8 km vs R72 100.4 km) → 72 h not separable (reading B: number of levels no worse). .
- Merged draft for the three lead times : model comparison 3/3 not no-worse; number of levels judged only at 24 h, at the margin. Sent for six-reviewer review (to rule on reading A/B and the stop category).
- 10-23 six-reviewer ruling: reading A (6/6). Finalized : model 3/3 not no-worse; number of levels (including the Q2 re-derivation) judged only at 24 h, at the margin; 48/72 h not separable. **Stop category: blocked, awaiting the PI** (A173 option ① fully run; option ② extend to ≥3 events, ③ AI model with full-level output, ④ invest no further, awaiting the PI).
- 11-12 22:04 A451: the lead recommends ④ invest no further: the model difference is 3/3 and the number of levels is secondary, so ③ does not address the cause; beyond 24 h both forcings have position errors >150 km, so adding events would most likely still not separate them; the wind-engineering bottleneck is gust magnitude (R24 maximum 10 m wind reaches only 0.55 of the observed gust), which is unrelated to the forcing fields. Awaiting the PI's decision.
- 11-12 22:07 the PI approved A451: ③ followed by ②. Dispatched r23-4 to do ③, the availability check of full-level AI models (read-only, 0 SU); the ② pre-registration will be written after ③ returns. Decision record: PI chose ③ followed by ② (20261112).
- 11-12 22:16 ③ check received: first choice GraphCast L37 (lead rechecked GCS weights 145990258 B; Brewer used L13), alternative NeuralGCM 0.7°. Lead decided: initialize from GFS analysis, use a common level table (GC37 ∩ actual GFS levels) to remove the confounding by number of levels, and drop the 13-level GFS group. Pre-check issued via the R23.4c conversation (≤15 CU); direct dispatch from the root conversation was wrongly blocked by pplan; reported to the platform.
- 11-13 09:01 GC37 pre-check received (R23.4c report): GFS 2020 has 33 levels, 31 levels in common; A100 single initialization ~76–100 s, 33.7 GB GPU memory; NaN = 0; drivers_gc37_v2 60/60; cumulative about

40 CU (cap within R23.4c raised from 15 to 50). Handed to R23.4c to pass to the WRF line for a smoke test and to draft the ② preregistration.

Key figures